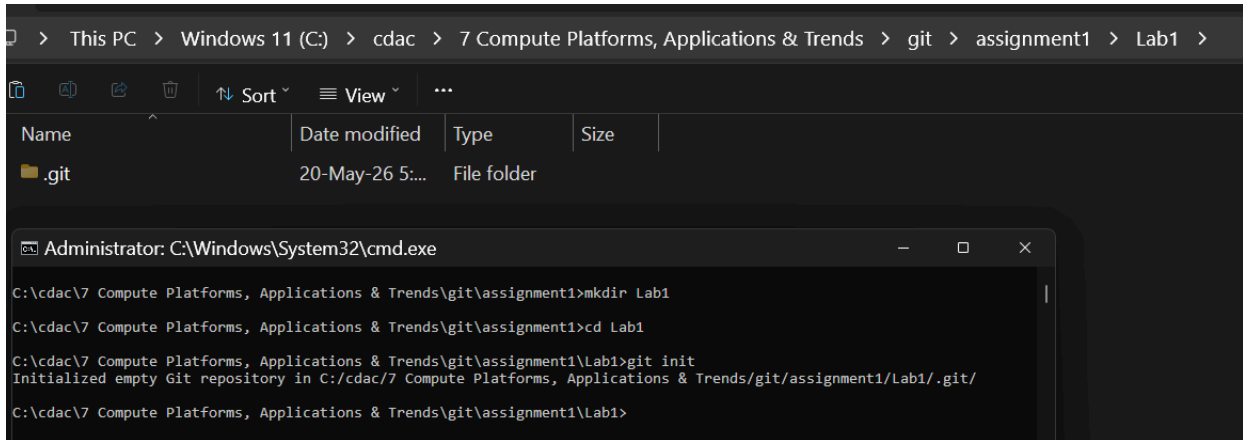


LAB-Assessment

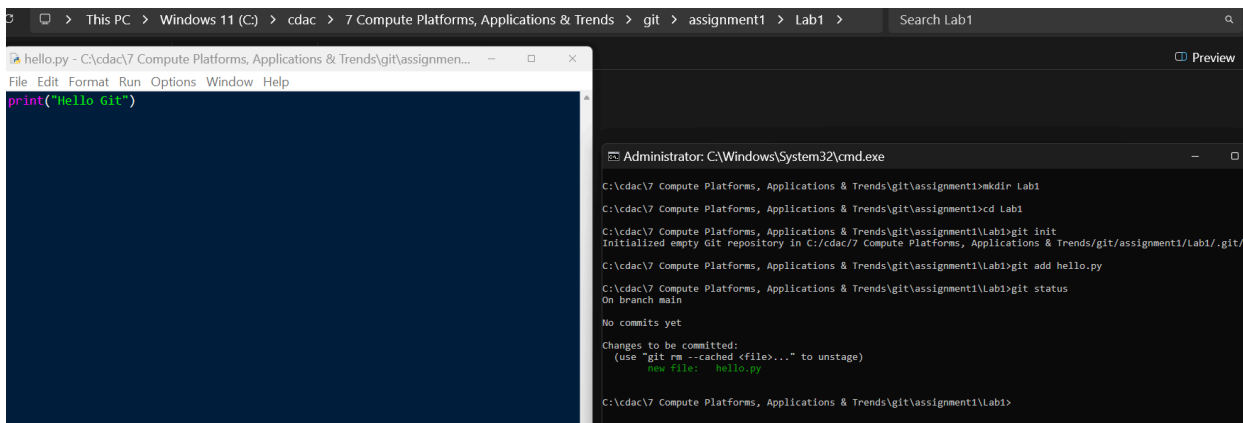
1. What is the command to initialize a new Git repository?



The screenshot shows a Windows File Explorer window with the address bar path: This PC > Windows 11 (C:) > cdac > 7 Compute Platforms, Applications & Trends > git > assignment1 > Lab1 >. The file list shows a folder named .git, modified on 20-May-26 5:00, and it is a File folder. Overlaid on this is a Command Prompt window titled "Administrator: C:\Windows\System32\cmd.exe". The command history shows the following commands and their outputs:

```
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1>mkdir Lab1
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1>cd Lab1
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git init
Initialized empty Git repository in C:/cdac/7 Compute Platforms, Applications & Trends/git/assignment1/Lab1/.git/
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>
```

2. What is the command to add files to the staging area?



The screenshot shows three windows. On the left is a text editor window titled "hello.py - C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1\hello.py" containing the code `print("Hello Git")`. In the middle is a File Explorer window showing the same directory as the first screenshot. On the right is a Command Prompt window titled "Administrator: C:\Windows\System32\cmd.exe" showing the following commands and outputs:

```
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1>mkdir Lab1
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1>cd Lab1
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git init
Initialized empty git repository in C:/cdac/7 Compute Platforms, Applications & Trends/git/assignment1/Lab1/.git/
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git add hello.py
On branch main
No commits yet
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   hello.py
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>
```

3. What is the command to commit changes with a message?

4. What is the command to check the status of the repository?

5. What is the command to create a new branch and switch to it?

6. What is the command to merge a branch into the current branch?

Merged feature branch into main

The screenshot shows a Windows file explorer window for the path `C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1`. It contains a `.git` folder and a `hello.py` file. Below it, a code editor shows the content of `hello.py`:

```
if __name__ == '__main__':  
    print("Hello Git")
```

To the right, a command prompt window shows the following commands and output:

```
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git switch feature  
Switched to branch 'feature'  
  
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git status  
On branch feature  
Changes not staged for commit:  
  (use "git add <file>..." to update what will be committed)  
  (use "git restore <file>..." to discard changes in working directory)  
        modified:   hello.py  
  
no changes added to commit (use "git add" and/or "git commit -a")  
  
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git add hello.py  
  
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git commit -m "modified hello"  
[feature ea87260] modified hello  
1 file changed, 2 insertions(+), 1 deletion(-)  
  
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git switch main  
Switched to branch 'main'  
  
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git merge feature  
Updating 8dd7fad..ea87260  
Fast-forward  
  hello.py | 3 ++-  
  1 file changed, 2 insertions(+), 1 deletion(-)  
  
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>
```

7. What is the command to view the differences between two branches or commits?

Comparing two commits

The screenshot shows a Windows file explorer window for the path `C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1`. It contains a `.git` folder and a `hello.py` file. Below it, a command prompt window shows the following commands and output:

```
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git log  
commit ea872608821c50c904e9f83da64866d965e7228a (HEAD -> main, feature)  
Author: Hrishikesh-2712 <chandrahrishikesh.2712@gmail.com>  
Date:   Wed May 20 17:34:24 2026 +0530  
  
    modified hello  
  
commit 8dd7fad75563b652fb6bab6a9d74cc8b15477279  
Author: Hrishikesh-2712 <chandrahrishikesh.2712@gmail.com>  
Date:   Wed May 20 17:23:51 2026 +0530  
  
    hello program created  
  
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git diff ea8726 8dd7fa  
diff --git a/hello.py b/hello.py  
index f14396c..078d6bf 100644  
--- a/hello.py  
+++ b/hello.py  
@@ -1,2 +1 @@  
-if __name__ == '__main__':  
-    print("Hello Git")  
+print("Hello Git")  
  
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>
```

Comparing(2 branch) main branch with feature after modification in main branch

hello.py - C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>

File Edit Format Run Options Window Help

```
def main():
    print("Hello Git")

if __name__ == '__main__':
    main()
```

```
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git diff feature..main
diff --git a/hello.py b/hello.py
index f14396c..ce61824 100644
--- a/hello.py
+++ b/hello.py
@@ -1,2 +1,5 @@
- if __name__ == '__main__':
+ def main():
+     print("Hello Git")
+
+ if __name__ == '__main__':
+     main()
```

8. What is the command to revert changes to a specific file?

before

The screenshot shows a code editor window with the following content:

```
hello.py - C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>
File Edit Format Run Options Window Help
def main():
    print("Hello Git, from CDAC")

if __name__ == '__main__':
    main()
```

Overlaid on the right is a terminal window titled "Administrator: C:\Windows\System32\cmd.exe". It shows the output of a git diff command:

```
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git diff feature..main
diff --git a/hello.py b/hello.py
index f14396c..ce61824 100644
--- a/hello.py
+++ b/hello.py
@@ -1,2 +1,5 @@
- if __name__ == '__main__':
+ def main():
+     print("Hello Git")
+
+ if __name__ == '__main__':
+     main()
```

Below the diff output, the terminal shows the status of the branch:

```
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   hello.py

no changes added to commit (use "git add" and/or "git commit -a")
```

after

```
hello.py - C:\cdac\7 Compute Platforms, Applications & Trends\git\assignmen...
File Edit Format Run Options Window Help
def main():
    print("Hello Git")

if __name__ == '__main__':
    main()

Administrator: C:\Windows\System32\cmd.exe
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git diff feature..main
diff --git a/hello.py b/hello.py
index f14396c..ce61824 100644
--- a/hello.py
+++ b/hello.py
@@ -1,2 +1,5 @@
-#if __name__ == '__main__':
+def main():
+    print("Hello Git")
+
+if __name__ == '__main__':
+    main()

C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   hello.py

no changes added to commit (use "git add" and/or "git commit -a")

C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git restore hello.py

C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>
```

9. What is the command to undo the last commit?

```
hello.py - C:\cdac\7 Compute Platforms, Applications & Trends\git\assignmen...
File Edit Format Run Options Window Help
<<<<<< HEAD
def main():
    print("Hello Git")

if __name__ == '__main__':
    main()

=====
print("Hello Git")
>>>>>> parent of ea87260 (modified hello)

Administrator: C:\Windows\System32\cmd.exe
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   hello.py

no changes added to commit (use "git add" and/or "git commit -a")

C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git restore hello.py

C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git restore HEAD~1
error: pathspec 'HEAD~1' did not match any file(s) known to git

C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git revert HEAD~1
Auto-merging hello.py
CONFLICT (content): Merge conflict in hello.py
error: could not revert ea87260... modified hello
hint: After resolving the conflicts, mark them with
hint: "git add/rm <paths>", then run
hint: "git revert --continue".
hint: You can instead skip this commit with "git revert --skip".
hint: To abort and get back to the state before "git revert",
hint: run "git revert --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
```

Encountered merge conflict

```
hello.py - C:\cdac\7 Compute Platforms, Applications & Trends\git\assignmen...
File Edit Format Run Options Window Help
print("Hello Git")

Administrator: C:\Windows\System32\cmd.exe
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git revert HEAD~1
Auto-merging hello.py
CONFLICT (content): Merge conflict in hello.py
error: could not revert ea87260... modified hello
hint: After resolving the conflicts, mark them with
hint: "git add/rm <paths>", then run
hint: "git revert --continue".
hint: You can instead skip this commit with "git revert --skip".
hint: To abort and get back to the state before "git revert",
hint: run "git revert --abort".
hint: Disable this message with "git config set advice.mergeConflict false"

C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git status
On branch main
You are currently reverting commit ea87260.
  (fix conflicts and run "git revert --continue")
  (use "git revert --skip" to skip this patch)
  (use "git revert --abort" to cancel the revert operation)

Unmerged paths:
  (use "git restore --staged <file>..." to unstage)
  (use "git add <file>..." to mark resolution)
        both modified:   hello.py

no changes added to commit (use "git add" and/or "git commit -a")

C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git add hello.py

C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git revert --continue
(main c5dfdbb) revert "modified hello"
1 file changed, 1 insertion(+), 5 deletions(-)

C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>
```

Resolved by keeping only print hello git

10. What is the command to push changes to a remote repository?

Before pushing I've to perform some pre-requisites

1.

So, here I renamed my branch to
feature/AI-07_Hrishikesh

```
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git branch -m AI-07_Hrishikesh
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git branch
* AI-07_Hrishikesh
  feature
```

Push to github

```
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git remote set-url origin https://github.com/adr
eshschauhan/remote-repo.git
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git branch
* AI-07_Hrishikesh
  feature
C:\cdac\7 Compute Platforms, Applications & Trends\git\assignment1\Lab1>git push origin AI-07_Hrishikesh_
```